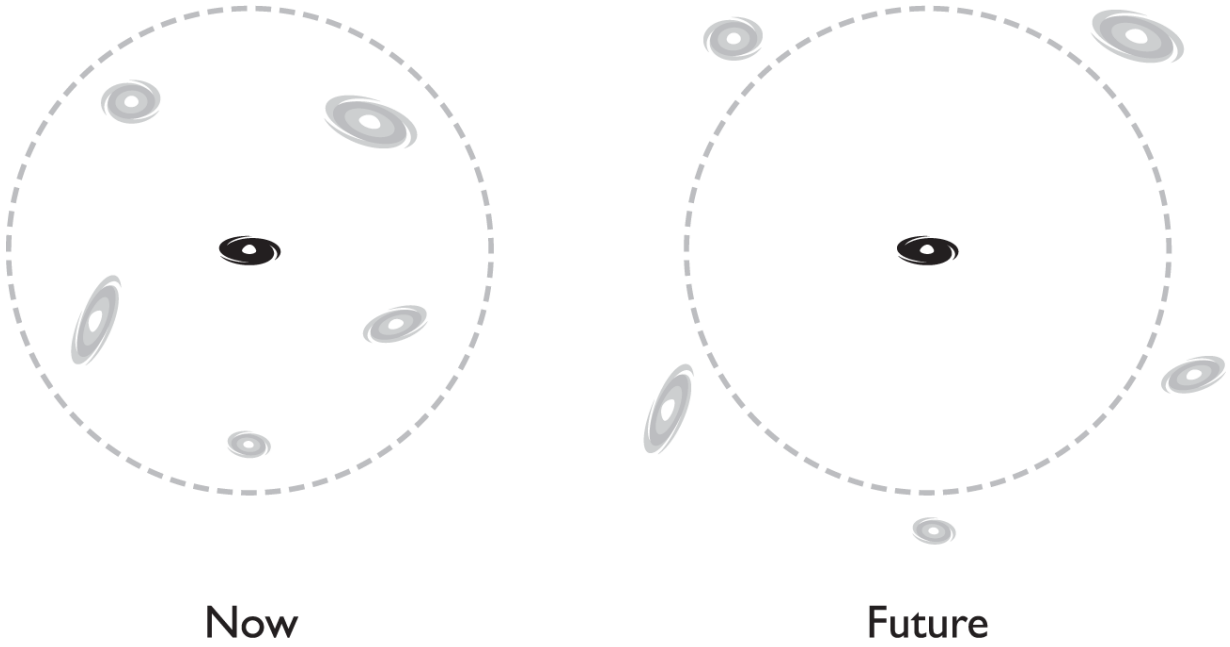


Figure 11: The Hubble radius now and in the future. As the expansion of the universe accelerates, galaxies that are currently inside our Hubble radius will be outside it. Eventually, no galaxies outside our Local Group will be visible.

Dark energy ruins everything, though. Because of dark energy, the expansion isn't slowing anymore—in fact, it has been speeding up for about the last five billion years. And while the Hubble radius is still technically growing in physical size, it's growing so slowly that the expansion is pulling previously visible objects outside it. We can see extremely distant objects whose light crossed into our Hubble radius before the acceleration began, but anything whose light isn't in the safe zone now will remain invisible forever. (More on that later.)

Even without the dark energy complication, an expanding universe can be a hard thing to wrap your head around.[fn8]

The fact that the universe is expanding means it was smaller in the past: fine.

The fact that it was smaller in the past means that something that is far away now was closer in the past: okay.

That, in turn, means that there's a very distant galaxy we can currently see that was, billions of years ago, kind of nearby: right.

And long ago that galaxy shot out a beam of light that was originally moving directly away from us despite being pointed in our direction, but which from our perspective then sort of stopped and turned around and has just now arrived: sure, from a certain point of view, that might make sense.